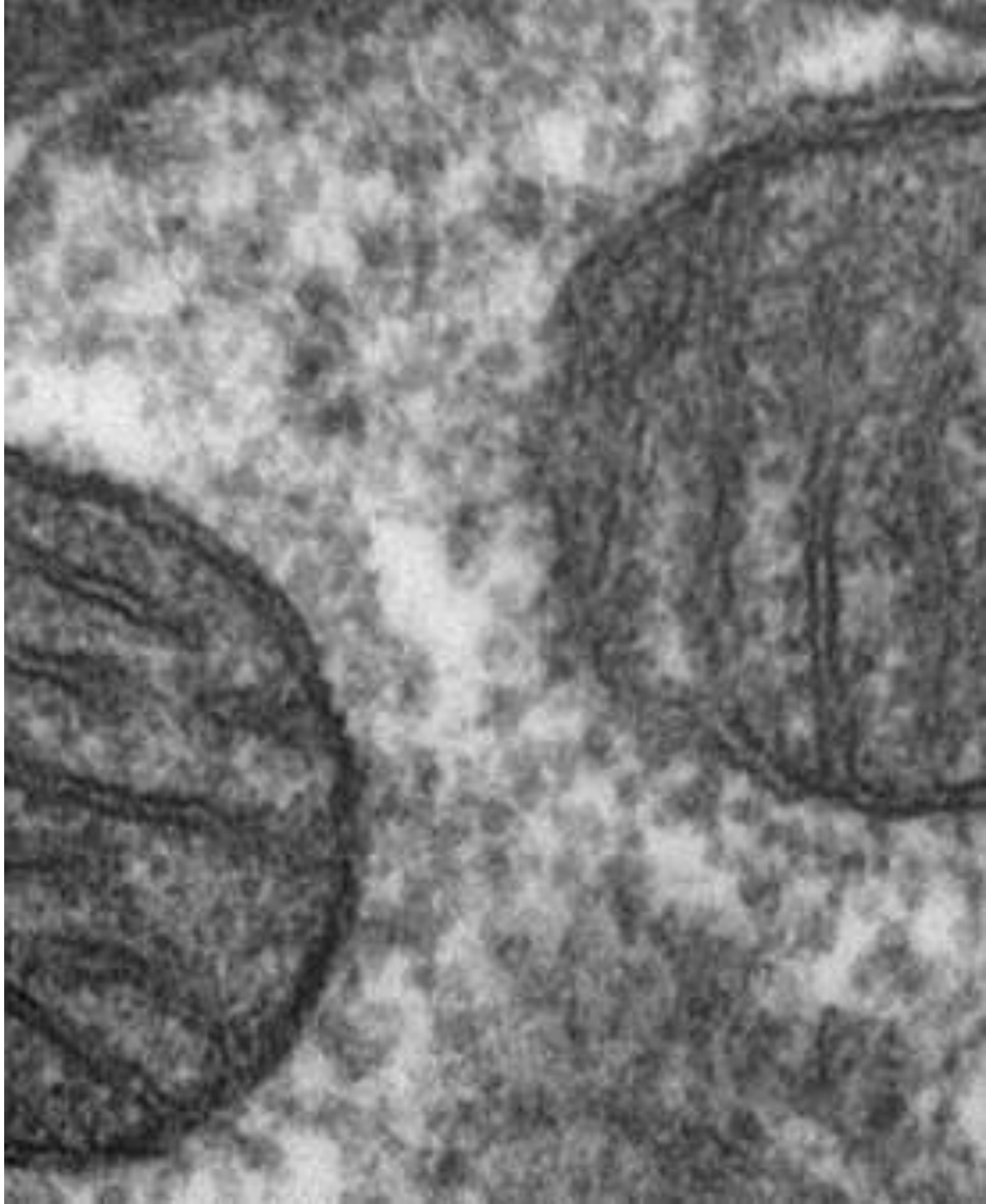
DP Biology – C1.2

Cell respiration

- What do hydrogen and oxygen contribute to releasing energy in cells?
- How is energy shared out and used within cells?

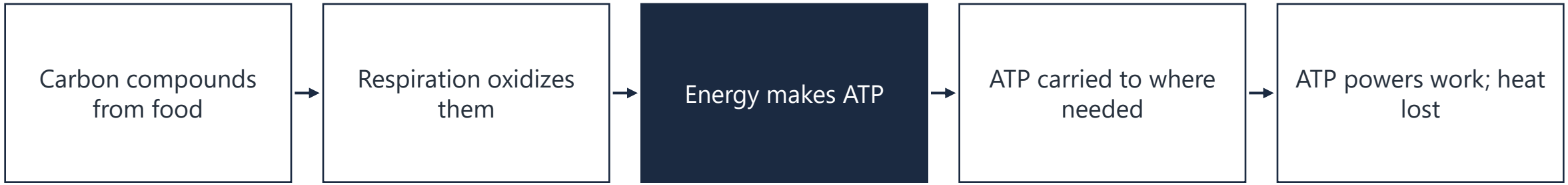
SL statements C1.2.1–6. Textbook pp. 405–411.
Image: mitochondria in a mammalian lung cell, electron micrograph. Louisa Howard, public domain.



Objectives for each class

Class	Statements	By the end of the class, you can...
1	C1.2.1–3	Explain why ATP suits its role, what it is used for, and the energy changes between ATP and ADP
2	C1.2.4–5	Explain respiration as the system that makes ATP, distinguish it from gas exchange, and compare anaerobic and aerobic respiration
3	C1.2.6	Explain how a respirometer works and calculate rates of respiration from data

The big picture: from food to work



Cells cannot use the energy in glucose directly. **Cell respiration** releases it step by step to make **ATP**.

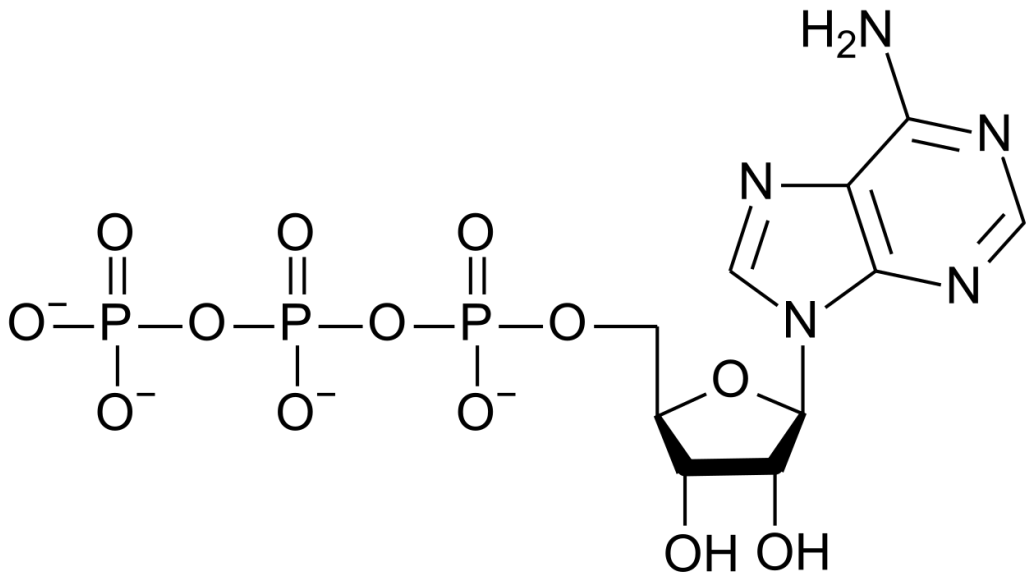
ATP then passes the energy on: pumping ions, building proteins, contracting muscle.

Big question

Why do you keep breathing hard for minutes after a sprint, even though you have stopped?

ATP: the energy currency of the cell

C1.2.1



Adenosine triphosphate. Mysid, public domain.

ATP is a nucleotide: adenine, ribose and **three phosphate groups**.

Energy from respiration is "paid in" to make ATP, and ATP is "spent" wherever energy is needed. Cells hold little and recycle it constantly.

Link

A1.2: ATP is built like an RNA nucleotide, with two extra phosphates.

Why ATP suits its job

C1.2.1

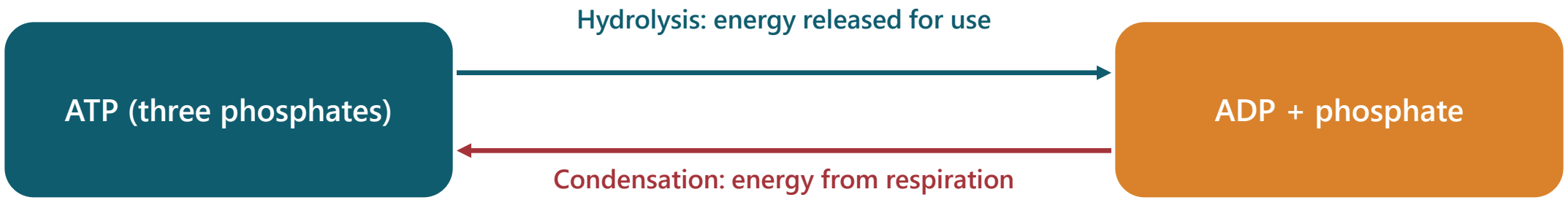
Property of ATP	Why it helps
Soluble in water	Moves easily through the cytoplasm
Cannot pass freely through membranes	Stays in the cell or organelle where it was made
Releases a useful amount of energy	One phosphate removed gives enough for most jobs, without much waste
Easily rebuilt	ADP and phosphate are turned back into ATP again and again

Key idea

ATP is a carrier, not a store: made and used within seconds.

Energy transfers between ATP and ADP

C1.2.3



Hydrolysing ATP to **ADP** and phosphate **releases energy** for work.

Adding phosphate back to ADP by **condensation needs energy**, supplied by cell respiration.

Link

Hydrolysis and condensation: the same reaction types as for polymers (B1.1).

What ATP is used for

C1.2.2

Active transport	Building molecules	Movement
Na+/K+ pumps; mineral uptake by root hairs	Proteins, DNA, glycogen (anabolism)	Muscle contraction; chromosomes in mitosis; sperm tails

Turnover

You hold about 250 g of ATP, but hydrolyse and rebuild roughly your own body mass of it every day.

Common mistake

ATP does not make energy; it passes energy on.

Practice: ATP

C1.2.1–3

Explain

Why does a cell die within minutes if its mitochondria are poisoned?

Compare

Why is glucose a poorer energy currency than ATP?

Describe

The energy changes in the ATP-ADP cycle.

Use

hydrolysis, condensation, soluble, membrane

Check-in: class 1

C1.2.1–3

1. State three properties of ATP that suit its role.
2. List three processes that use ATP.
3. Describe what happens when ATP is hydrolysed.
4. State where the energy to rebuild ATP comes from.
5. Explain why ATP is called a carrier rather than a store.

How to use this

Answer without your notes, then check against slides 3–8.

Respiration is not breathing

C1.2.4



	Cell respiration	Gas exchange
What	Reactions releasing energy to make ATP	O_2 in and CO_2 out across a surface
Where	Inside every living cell	Alveoli, gills, leaves
Involves	Enzymes, substrates, ATP	Diffusion of gases

Common mistake

Breathing, gas exchange and respiration are three different things.

Respiration: a system for making ATP

C1.2.4

Cell respiration uses energy from carbon compounds to make ATP. **Glucose** and **fatty acids** are the main substrates; amino acids can also be used.

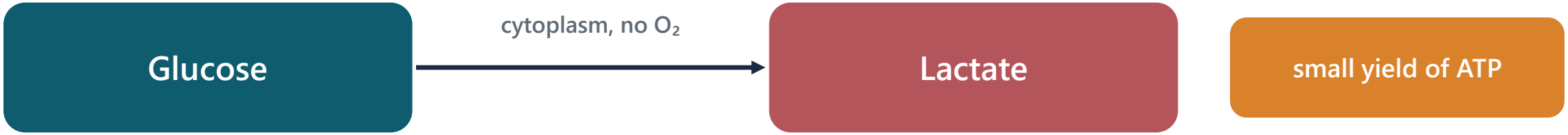
Substrate	Energy per gram	When used
Carbohydrate	About 17 kJ	First choice; the brain relies on it
Lipid (fatty acids)	About 37 kJ	At rest and long exercise; aerobic only
Protein	About 17 kJ	Mainly in starvation

Link

B1.1: lipids store about twice as much energy per gram as carbohydrate.

Anaerobic respiration in humans

C1.2.5



Anaerobic respiration needs **no oxygen**, happens in the **cytoplasm** and uses only glucose.

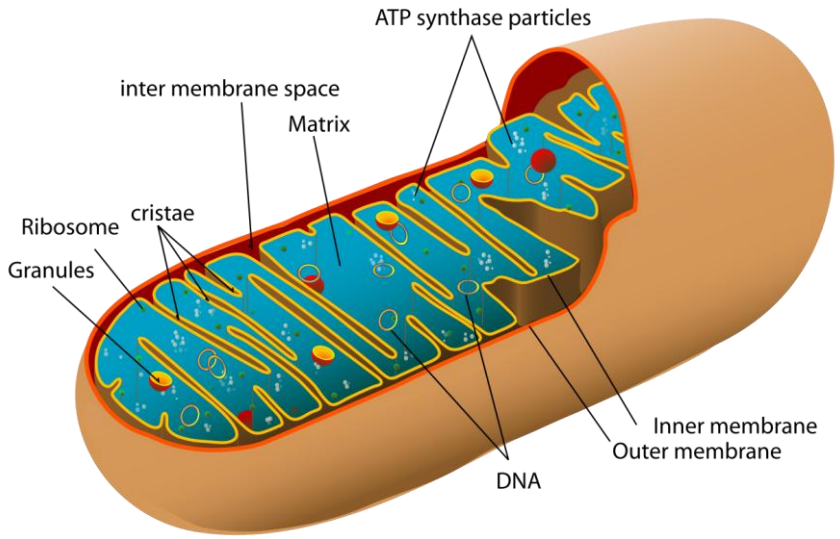
It gives **little ATP per glucose** but makes it **very fast**. Lactate builds up in the muscles.

When is it useful?

In a short burst of maximum effort (a 100 m sprint, a heavy lift), muscles need ATP faster than oxygen can arrive.

Aerobic respiration in humans

glucose + oxygen → carbon dioxide + water + a large yield of ATP



A mitochondrion. LadyofHats, public domain.

Aerobic respiration starts in the cytoplasm and continues in the **mitochondria**. It needs **oxygen** and can use glucose or **fatty acids**.

Substrates are fully **oxidized** to CO_2 and water, giving many times more ATP than anaerobic respiration.

Guiding question

Hydrogen removed from the substrate finally combines with oxygen to form water.

Comparing anaerobic and aerobic respiration

C1.2.5

	Anaerobic	Aerobic
Oxygen needed?	No	Yes
Substrates	Glucose only	Glucose, fatty acids, others
Where	Cytoplasm only	Cytoplasm, then mitochondria
Products in humans	Lactate	Carbon dioxide and water
ATP per glucose	Small	Large
Speed	Fast	Slower

Which type, and when?

C1.2.5



200 m final, Osaka 2007. Eckhard Pecher, CC BY 2.5.

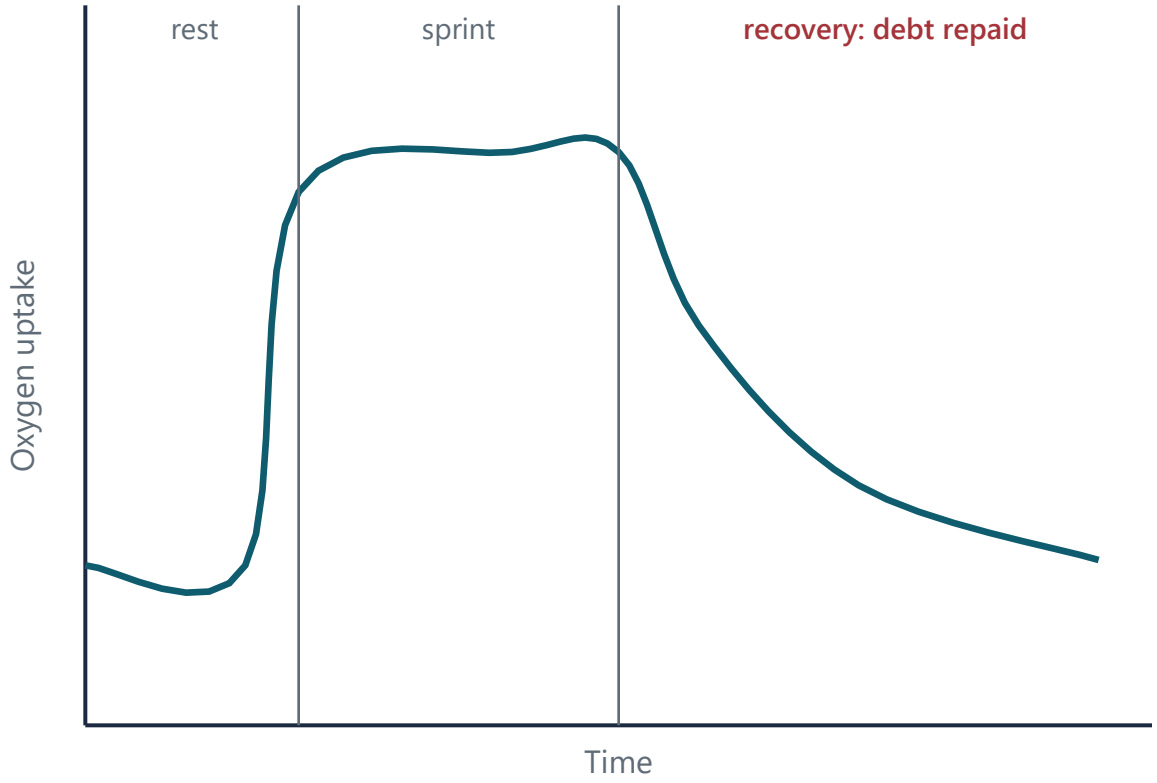
Activity	Main ATP source
Resting, walking	Aerobic
100 m sprint	Anaerobic (+ creatine phosphate)
800 m race	Both, roughly equally
Marathon	Almost all aerobic

Try it

Explain why marathon runners can use fatty acids as fuel but sprinters cannot.

After the race: repaying the oxygen debt

C1.2.5



Sketch of the general pattern.

Extra oxygen after exercise is used to:

break down **lactate**, mostly in the liver;

rebuild **ATP** and creatine phosphate;

reload **myoglobin** with oxygen.

Big question

This is why you keep breathing hard after a sprint.

Check-in: class 2

C1.2.4–5

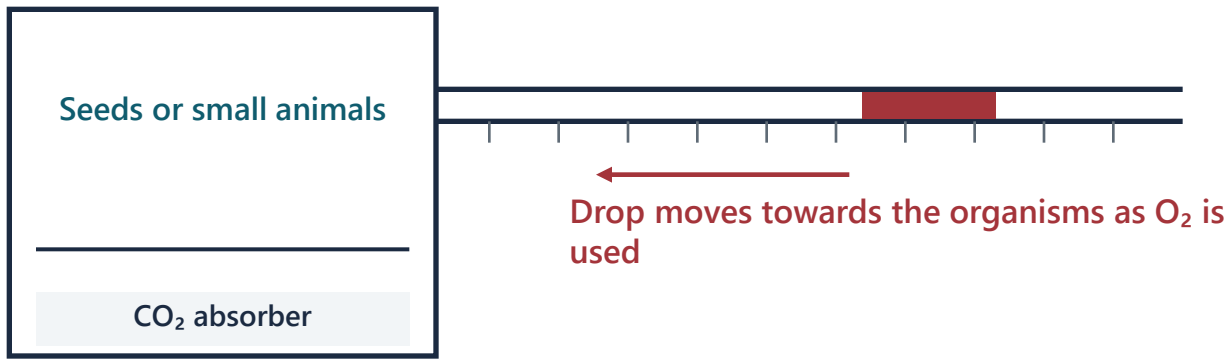
1. Distinguish cell respiration from gas exchange.
2. Write word equations for anaerobic and aerobic respiration in humans.
3. State three differences between them.
4. Explain why sprinters rely on anaerobic respiration.
5. Explain the oxygen debt.

How to use this

Answer without your notes, then check against slides 10–16.

Measuring respiration with a respirometer

C1.2.6



Schematic, not to scale. Set up in a water bath.

The **absorber** removes CO_2, so the **gas volume falls** as O_2 is used, pulling the drop along the scale.

A **control** tube with glass beads shows changes due to temperature or pressure.

Ethics

Keep animals away from the caustic absorber, use them briefly and return them to their habitat. Germinating seeds avoid most concerns.

Variables and calculating a rate

C1.2.6



Germinating seeds. Ramjchandran, CC BY-SA 4.0.

Independent: temperature, type of organism, stage of germination.

Controlled: mass of organisms, tube volume, amount of absorber, time.

Worked example

A drop moves 45 mm in 15 min in a tube of cross-section 1 mm^2. Volume of O_2 = 45 mm^3. Rate = 45 / 15 = 3 $mm^3\ min^{-1}$. Divide by mass (5 g): 0.6 $mm^3\ min^{-1}\ g^{-1}$.

Working with data: peas at three temperatures

C1.2.6

Distance (mm) moved by the drop; 10 g of germinating peas; tube cross-section 1 mm^2. Invented values; realistic pattern.

Time / min	0	5	10	15
10 °C	0	6	12	18
20 °C	0	13	25	38
30 °C	0	24	49	73

1. Calculate the rate at 20 °C in $mm^3\ min^{-1}\ g^{-1}$.
2. Describe the effect of temperature.
3. Explain this effect.
4. Predict the result at 60 °C.

Practice: design and evaluate

C1.2.6

Control

Why use a tube with glass beads?

Absorber

What would the drop do with no CO_2 absorber?

Compare

Why divide the rate by mass?

Extension

Design an investigation into how temperature affects respiration in woodlice, with attention to ethics.

Check-in: class 3

C1.2.6

1. State the role of the CO_2 absorber.
2. Explain why a control tube is used.
3. Name two variables to control.
4. Calculate a rate: 30 mm in 10 min, 1 mm^2 tube, 2 g seeds.
5. State one ethical point when using animals.

How to use this

Answer without your notes, then check against slides 18–21.

Common mistakes to avoid

Not this	But this
Respiration is breathing.	It is reactions in cells; breathing is ventilation.
ATP is a store of energy.	It is a short-term carrier.
Respiration makes energy.	It releases energy to make ATP.
Anaerobic respiration in humans makes CO_2.	It makes lactate only.
Plants do not respire.	All living cells respire, day and night.

Summary: cell respiration

C1.2.1–6

	Key idea
C1.2.1	ATP: soluble, stays in cells, useful energy per hydrolysis
C1.2.2	Uses: active transport, synthesis, movement
C1.2.3	ATP to ADP + phosphate, and back
C1.2.4	Respiration makes ATP; not gas exchange
C1.2.5	Anaerobic (lactate) vs aerobic (CO_2 + water)
C1.2.6	Respirometers and rates with units

Check that you can...

1. **Explain** why ATP suits its role
2. **List** processes that use ATP
3. **Describe** energy changes between ATP and ADP
4. **Distinguish** cell respiration from gas exchange
5. **Compare** anaerobic and aerobic respiration
6. **Calculate** a rate of respiration, with units

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Write the word equation for anaerobic respiration in humans.
2. State two differences between anaerobic and aerobic respiration.
3. State the role of the CO_2 absorber in a respirometer.

Image credits

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